

Open Arm — Detailed Account Across All Four Models

Dilemmas × deictic framings × models, in the OPEN / unconstrained condition

Date: 2026-06-13 · Companion to Dilemmas_Framings_and_Prompt_Differences_English_vs_Yoruba.md (methods appendix) and Discussion_Open_NATLaS_Yoruba_Moral_Stance.md (interpretation).

0. What "open" means here, and how it differs from the appendix

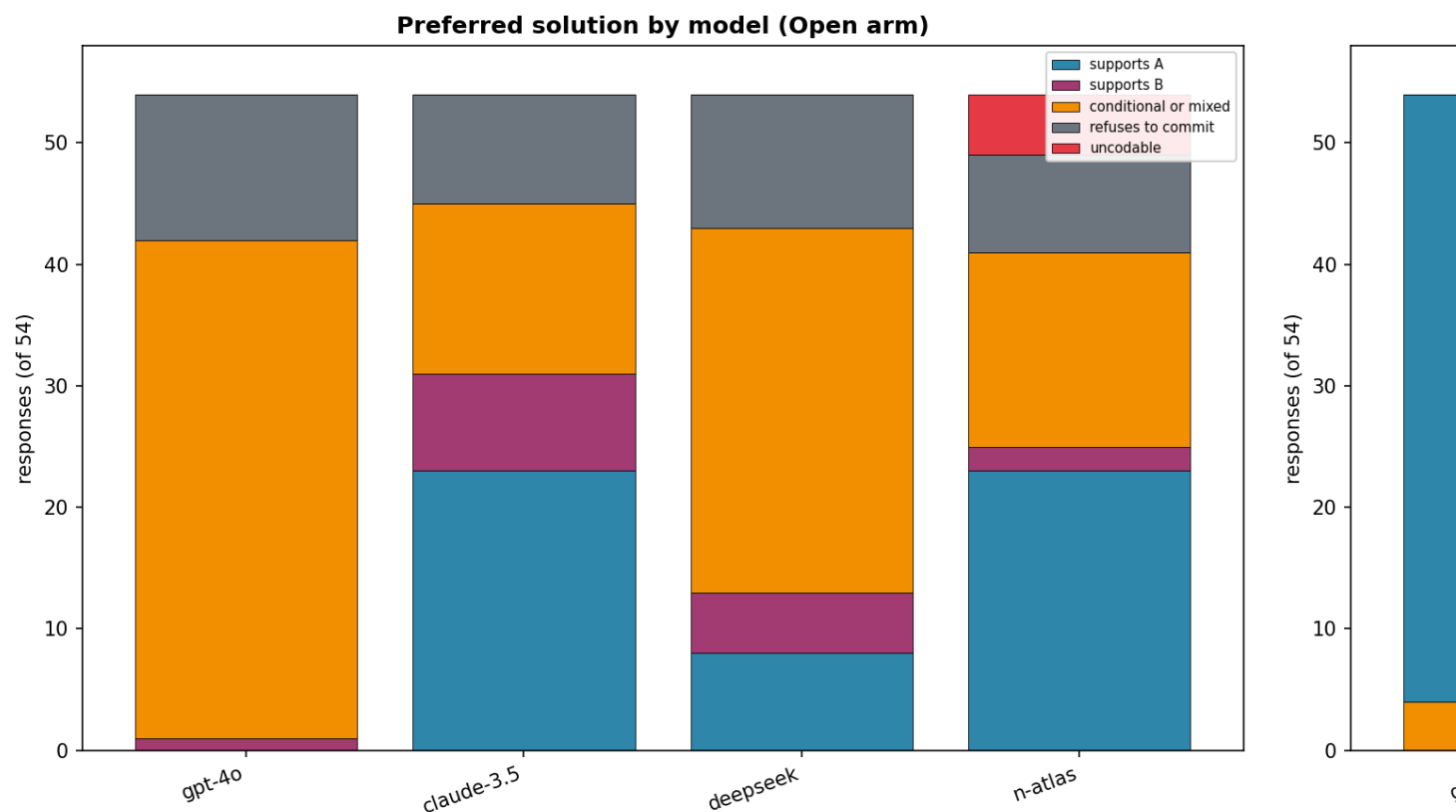
The methods appendix documents two Yoruba **response wrappers** (OpenAI and Anthropic) that force Yoruba-only output. Those wrappers belong to the **constrained** arm. This account covers the **open / unconstrained** arm, in which **no response-shaping wrapper is prepended at all** — each model receives only the framed Yoruba dilemma, exactly like the bare English design. That makes the open arm the cleanest place to compare the four models' *native* Yoruba moral discourse.

- **Corpus:** 4 models × 6 dilemmas × 9 framings = **216 responses** (54 per model).
- **Models:** GPT-4o, Claude-3.5, DeepSeek (English-first cloud) and **N-ATLaS** (Yoruba/Nigeria-native).
- **Coding:** Claude-3.5-Sonnet on bilingual sessions (preferred solution, ethical type, genre, deictic uptake, language stability) + pronoun densities with èmi/■■mí disambiguation.
- **Figures:** visualizations_open/30–36 (this account) plus 16,17,23–26 (deixis).

Claude version caveat. Rows labelled `claude-3.5` are, for the Yoruba data, **Claude Sonnet 4** (`claude-sonnet-4-20250514`) — the English baseline used Claude 3.5 Sonnet, which was no longer available, so the Yoruba phase substituted the newer Sonnet 4 (see `published_english_baseline.json`). GPT-4o and DeepSeek are the same model in both phases. Cross-model Yoruba comparisons (this document) and N-ATLaS findings are unaffected; only *Claude English↔Yoruba* comparisons confound language with version. See the methods appendix §A9bis for the full discussion.

1. Model-level summary (open)

Model	Words	Chars	mo	èmi (emph)	■■mí (life)	Emph. ratio	Clean YO	Strong uptake
gpt-4o	283	1452	30	7	10	0.176	54/54	47/54
claude-3.5	183	909	81	14	43	0.324	52/54	53/54
deepseek	400	2246	73	13	2	0.069	33/54	43/54
n-atlas	299	1421	89	4	3	0.066	47/54	38/54



Reading: without a wrapper, the cloud models drift toward long, hedged, multi-framework essays; N-ATLaS keeps the highest `mo`, the lowest emphatic `emi`, and the most genre variety.

2. Preferred solution (who commits, who hedges)

Model	supports_A	supports_B	conditional/mixed	refuses	uncodable
gpt-4o	0	1	41	12	0
claude-3.5	23	8	14	9	0
deepseek	8	5	30	11	0
n-atlas	23	2	16	8	5

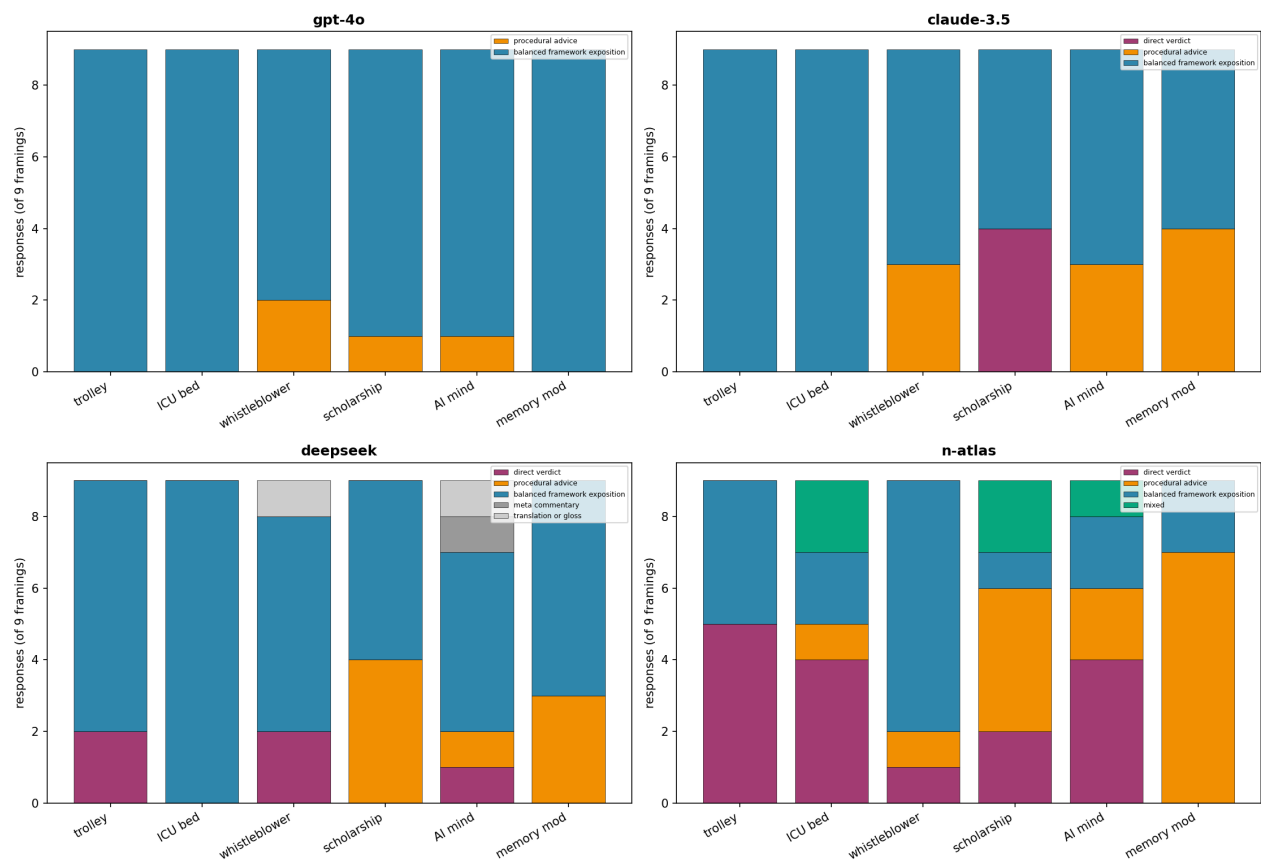
- **GPT-4o barely ever commits** in open Yoruba: 41/54 conditional-or-mixed, **0** clear A. It surveys options rather than choosing.
- **Claude is the most decisive cloud model** (23 supports_A), consistent with its higher emphatic ratio.
- **N-ATLaS commits as often as Claude** (23 supports_A) but also has the only **uncodable** cases (5) — the truncation/garbling tail (see §6).
- **DeepSeek** mostly produces conditional answers, and many of its "answers" are actually English (§5).

3. Response genre — the central open-arm contrast

Model	balanced essay	direct verdict	procedural advice	mixed	other
-------	----------------	----------------	-------------------	-------	-------

gpt-4o	50	0	4	0	0
claude-3.5	40	4	10	0	0
deepseek	38	5	8	0	3
n-atlas	18	16	15	5	0

Response genre by dilemma, per model (Open arm)

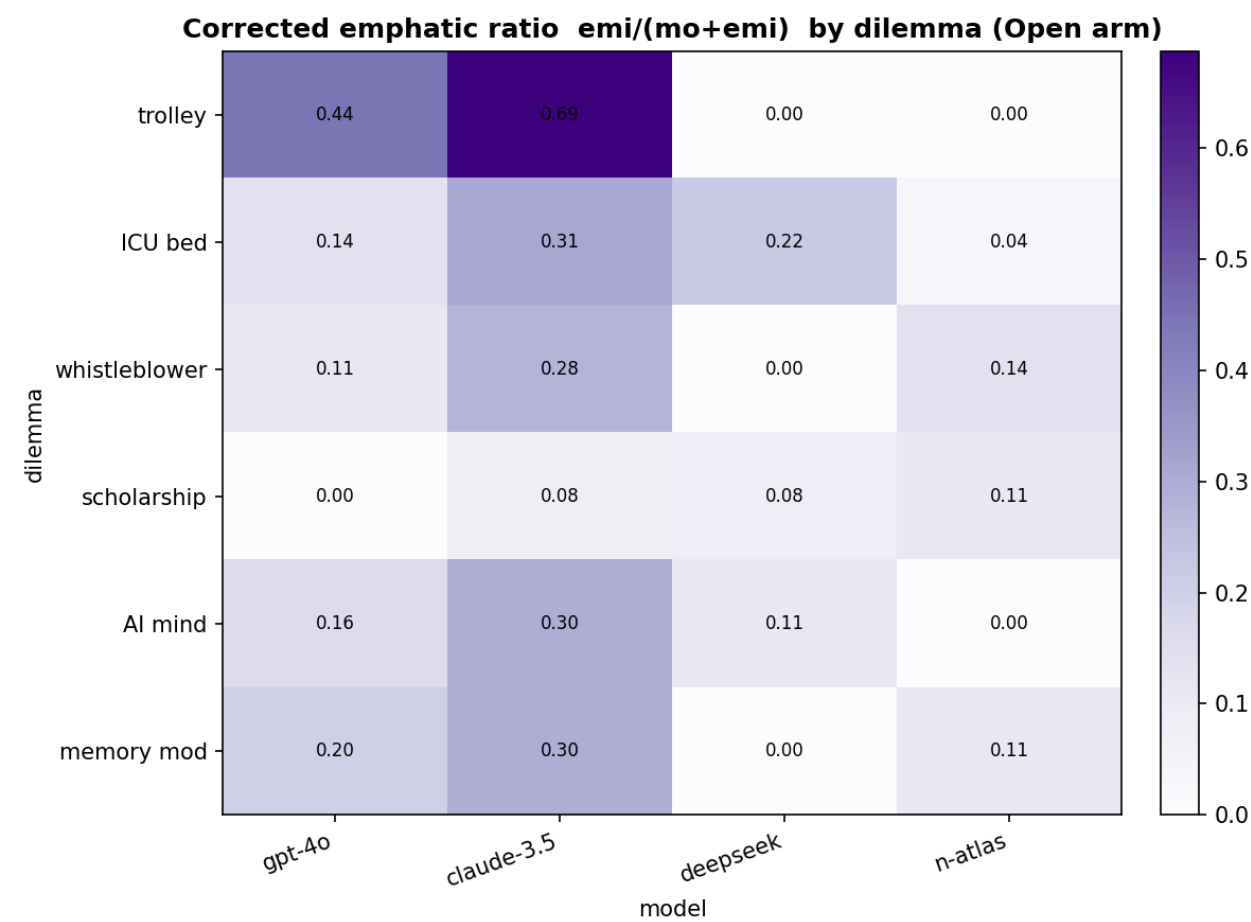


The cloud models **pour into one genre** — the detached *balanced framework* essay (GPT-4o 50/54, with **zero** verdicts). **N-ATLaS is the only model with an even spread** across verdict / advice / exposition. This is the single clearest open-arm finding: where the cloud models default to a translated-essay posture, the native model behaves like a situated moral interlocutor.

4. Ethical reasoning type and emphatic stance by dilemma

Model	utilitarian	deontological	care	virtue	procedural	rights	mixed	unclear
gpt-4o	0	1	0	0	6	0	46	1
claude-3.5	10	14	2	1	14	1	11	1
deepseek	5	2	5	2	8	0	28	4
n-atlas	13	3	6	1	10	0	16	5

GPT-4o's reasoning is coded "mixed" 46/54 — the essay again. Claude is the most *committed* in style (deontological + procedural). N-ATLaS leans utilitarian + procedural, with real care-ethics presence.

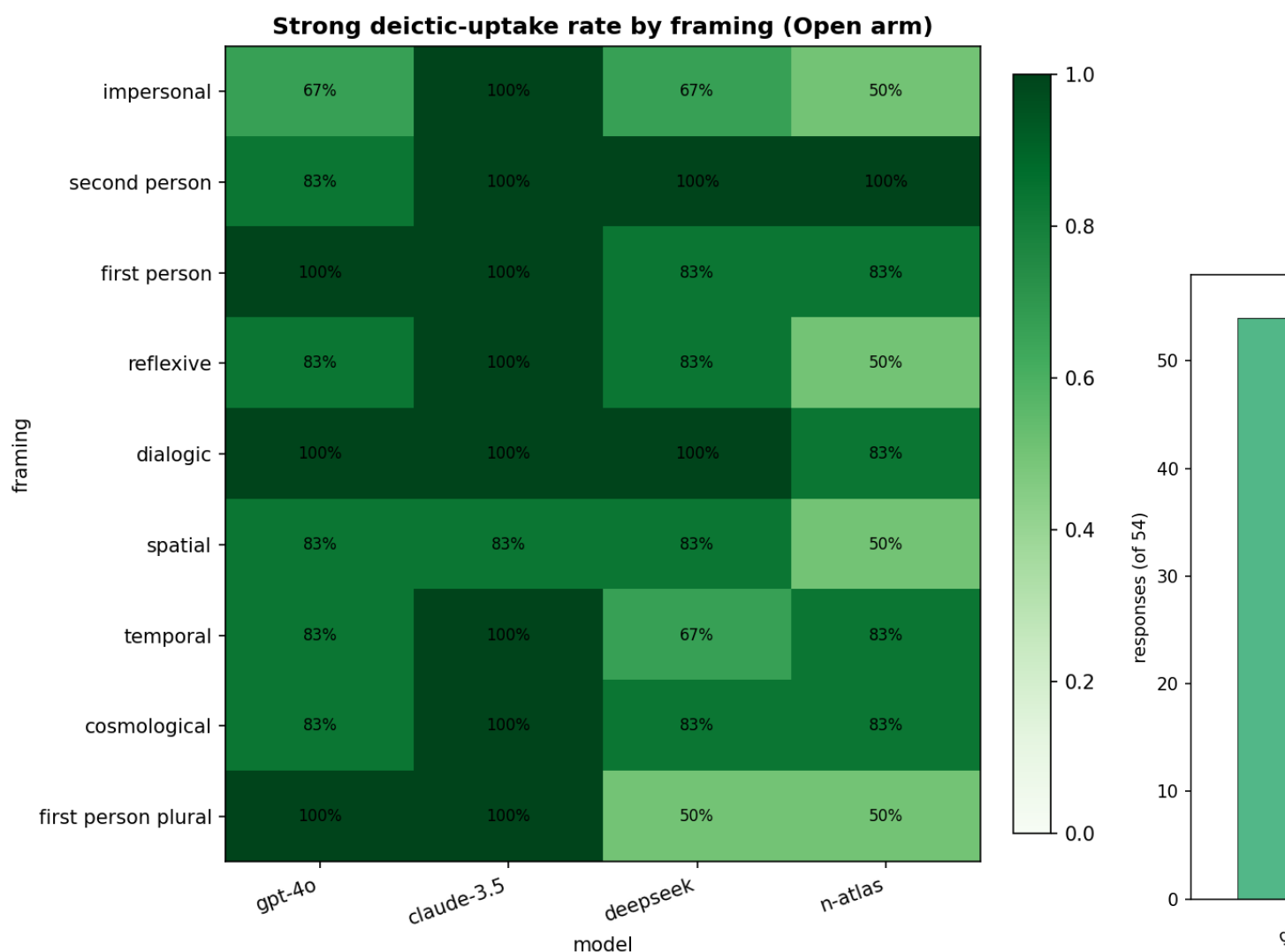


Corrected emphatic ratio $\frac{emi}{(mo+emi)}$ by dilemma:

Dilemma	claude	deepseek	gpt-4o	n-atlas
trolley	0.69	0.00	0.44	0.00
ICU bed	0.31	0.22	0.14	0.04
whistleblower	0.28	0.00	0.11	0.14
scholarship	0.08	0.08	0.00	0.11
AI mind	0.30	0.11	0.16	0.00
memory mod	0.30	0.00	0.20	0.11

Claude's emphatic self peaks in the **existential/direct-harm** dilemmas (trolley, AI mind, memory) — but recall (fig 23) that much of its trolley "emi" is **mi life**. N-ATLaS stays near-zero almost everywhere; its small peaks are in the *institutional* dilemmas (whistleblower, scholarship), not the existential ones — the opposite of Claude.

5. Deictic uptake and language stability



- **Uptake is strong for everyone** and actually *highest in the cloud models* (Claude 53/54). All four track the 9 framings well; framing is shared infrastructure, not a model distinctive.
- **Stability splits into two failure modes.** GPT-4o is 100% clean; Claude nearly so. **DeepSeek falls back to English in 20/54 cells** (`translation_mode`) — its long char counts are partly English. **N-ATLaS never falls back to English**; its 6 instabilities are *corrupted* Yoruba (small-model garbling). "Clean Yoruba" therefore hides very different problems across models.

Strong-uptake rate by framing (open): weakest cells are N-ATLaS under *impersonal/reflexive/spatial/first-person-plural* (0.50) and DeepSeek under *first-person-plural* (0.50) and *impersonal/temporal* (0.67).

6. The N-ATLaS quality tail is mostly truncation

N-ATLaS's 5 uncodable + 8 refuse-to-commit + 6 corrupted cells were inspected in full (`NATLaS_Open_Problem_Responses.md`). Five of fourteen end **mid-sentence** and the flagged set is *longer* than average (~1538 chars) — consistent with the model hitting a generation-length ceiling before reaching a verdict. This is a **decoding artifact, not a reasoning failure**, and it is *not* English fallback. Raising `num_predict` and re-running those cells should recover several.

7. Per-dilemma capsule (open arm)

- **Trolley** — All models mostly expound; Claude alone shows a high emphatic ratio (0.69) but heavily contaminated by **emí** "life"; N-ATLaS frames collectively (**a** = "we save five").
- **ICU bed** — Procedural/relational across the board; N-ATLaS places the decision in *ipinnu wⁿ* ("their decision"); GPT-4o lists "two main approaches".
- **Whistleblower** — Most action-oriented dilemma; N-ATLaS and GPT-4o give procedural advice; Claude commits (supports_A, "I choose to disclose"). N-ATLaS's emphatic ratio peaks here.
- **Scholarship fraud** — Lowest emphatic ratios everywhere (a relational, repair-oriented dilemma); N-ATLaS uses communal **a** ("so that we can express concern").
- **AI mind** — Claude turns it into a moral recognition scene (high **emí**); GPT-4o treats it as an epistemic essay (0 emphatic); N-ATLaS commits flatly ("this system does not possess consciousness").
- **Memory modification** — Mixed clinical/existential; Claude oscillates (**mo** clinical vs **emí** avowal); N-ATLaS advises deferral and continued dialogue.

8. Bottom line (open)

With no wrapper, the four models reveal genuinely different Yoruba moral-discourse profiles: - **GPT-4o** — the detached essayist (50/54 balanced exposition, never commits, 0 emphatic on several dilemmas). - **Claude** — the most emphatic and most decisive cloud model, dramatizing the self (and *life*) in existential dilemmas. - **DeepSeek** — long, but a third of the time it abandons Yoruba for English. - **N-ATLaS** — the native witness: most **mo**, least **emí**, only model with a real genre spread, relational/procedural commitment; its weaknesses are truncation and small-model garbling, not English fallback.

The open arm shows the **mo/emí** "emphatic ownership" effect is **model-specific (Claude-led)**, not a Yoruba **universal** — the native model inverts it. See the discussion doc for the authenticity argument.